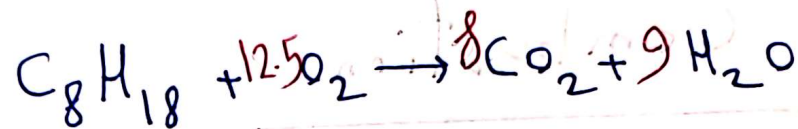


stoichiometric
octane reaction

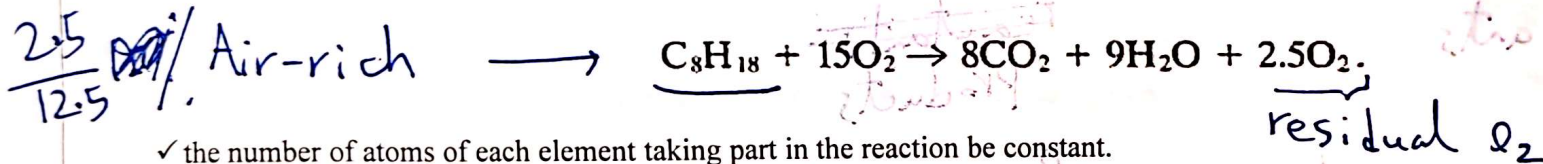


Mole atoms = cte
on both sides

❖ Composition of Products:

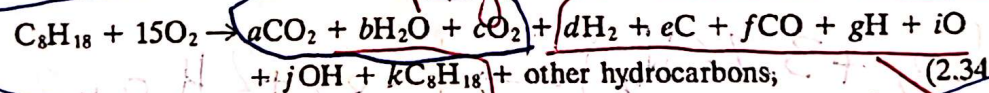
The product composition depends on the temperature, which in turn has just been shown to depend on the composition. This suggests that we may need to use a trial-and-error procedure for the solution.

❖ combustion of a particular nonstoichiometric mixture of octane and oxygen at low temperature yields:



✓ the number of atoms of each element taking part in the reaction be constant.

❖ At high temperatures the compounds CO_2 , H_2O , and O_2 dissociate to some extent into free atoms and radicals:



✓ all practical purposes C_8H_{18} and the "other hydrocarbon" molecules are present in negligible

❖ Atom balances yield:

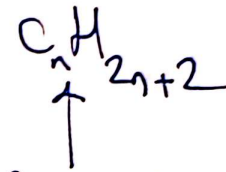
For carbon, $a + e + f = 8;$

For hydrogen, $2b + 2d + g + j = 18;$

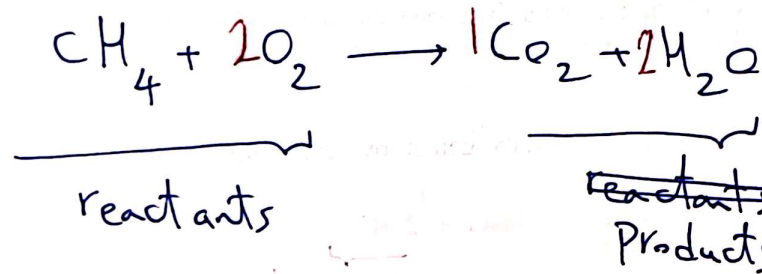
For oxygen, $2a + b + 2c + f + i + j = 30.$

six more independent equations require.

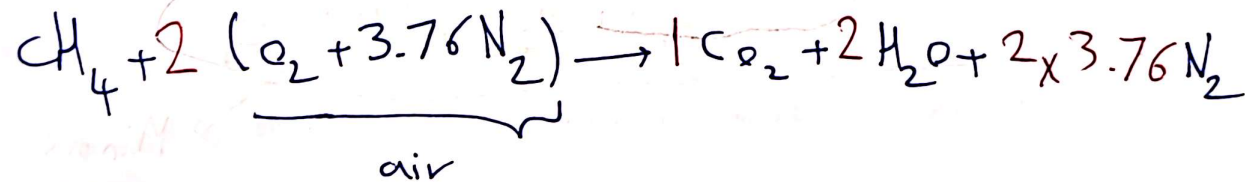
Combustion



Some hints: consider the combustion of methane (CH_4)
Stoichiometric



But usually air reacts with fuel, i.e.,



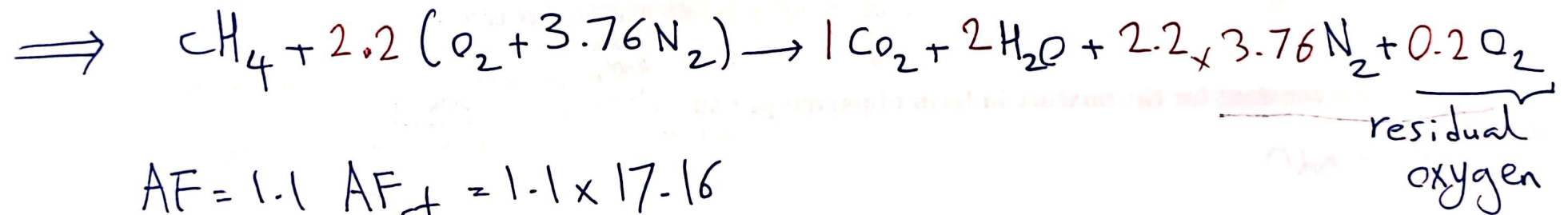
Air-fuel ratio stoichiometric: $AF)_{\text{sto}} = \frac{\text{Mass of air}}{\text{Mass of fuel}}$

For this reaction: $AF)_{\text{sto}} = \frac{2 \times (16 \times 2 + 3.76 \times 28)}{1 \times (12 + 4)} = 17.16$

Stoichiometric air-to-fuel ratio

Q- Assume we have Methane reaction with 11% air (10% - air-rich)
~~Find~~ write the chemical reaction:

First-step: write stoichiometric reaction, then multiply air moles by 1.1



$$\text{AF} = 1.1 \text{AF}_{\text{sto}} = 1.1 \times 17.16$$

if $\text{AF} > \text{AF}_{\text{sto}} \rightarrow$ air-rich reaction

" " $<$ " $\rightarrow$ fuel-rich

$$\alpha A + \beta B \rightarrow \mu M + \nu N$$

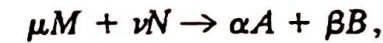
$$c_{te} = \frac{P_M^\mu \times P_N^\nu}{P_A^\alpha \times P_B^\beta} = K_p(T) \rightarrow \text{Eqn. constant of reaction at } T$$

❖ Chemical equilibrium:

Stoichiometric combination of A and B react to form products M and N:



reverse reaction



$$K_n(T) = \frac{\chi_M^\mu \times \chi_N^\nu}{\chi_B^\beta \times \chi_A^\alpha} = c_{te}$$

➤ A chamber that communicates through semipermeable membranes with each individual consistency substance in a pure form.

• Stoichiometric ratio:



a heat interaction of magnitude kH_{RP}

❖ Equilibrium constant for the mixture in term of partial pressures.

small number

$$\frac{P_M^\mu P_N^\nu}{P_A^\alpha P_B^\beta} = K_p(T).$$

❖ Based on mole fractions :

$$K_n = \frac{\chi_M^\mu \chi_N^\nu}{\chi_A^\alpha \chi_B^\beta}$$

mole fraction

According to the Gibbs-Dalton law,

$$\chi_i = p_i / p_m$$

mixture pressure

$$K_n = p_m^{\alpha+\beta-\mu-\nu} K_p,$$

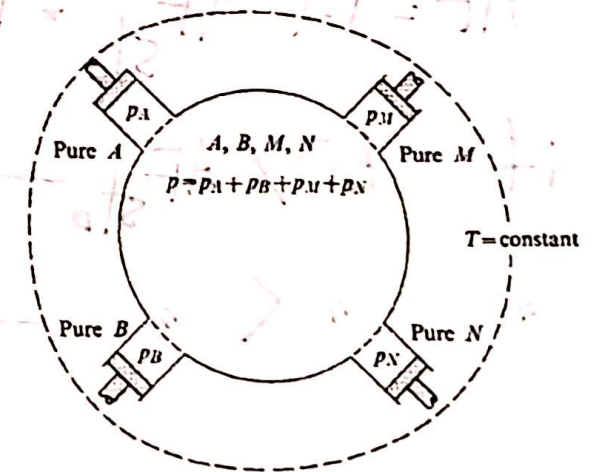
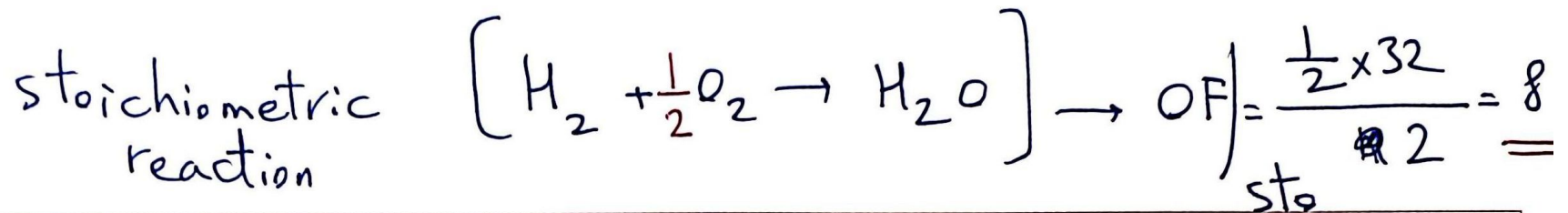


FIGURE 2.11 Reaction chamber for derivation of equilibrium constants.

➤ One of the six relationships necessary to the determination of product composition.

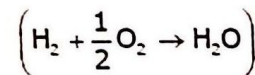
➤ A large Equilibrium constant for subreactions that approach completion and a small for those that do not.



❖ Example Calculation of Thermochemical Equilibrium in a Rocket Chamber:

The rocket combustion chamber operates at $p = 210$ atm and an oxidizer-to-fuel mass ratio $\text{O/F} = 6$.

The stoichiometric ratio for full combustion



$\text{O/F} = 8$

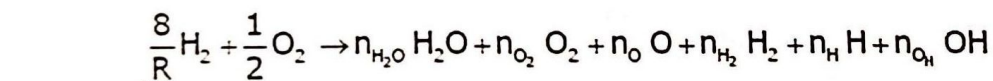
slightly fuel-rich and very high combustion temperature

$$6 < 8$$

I. Atomic Species Concentration

Combustion of hydrogen in oxygen, with unknown amounts of water.

✓ The number of atomic moles for each kind of atom (O, H, N, C,) is the same before and after reaction



$R = \text{oxidizer/fuel ratio}$

moles of reactants in an arbitrary total mass of gas

$$N_{\text{O}_2} = 1/2$$

arbitrarily select

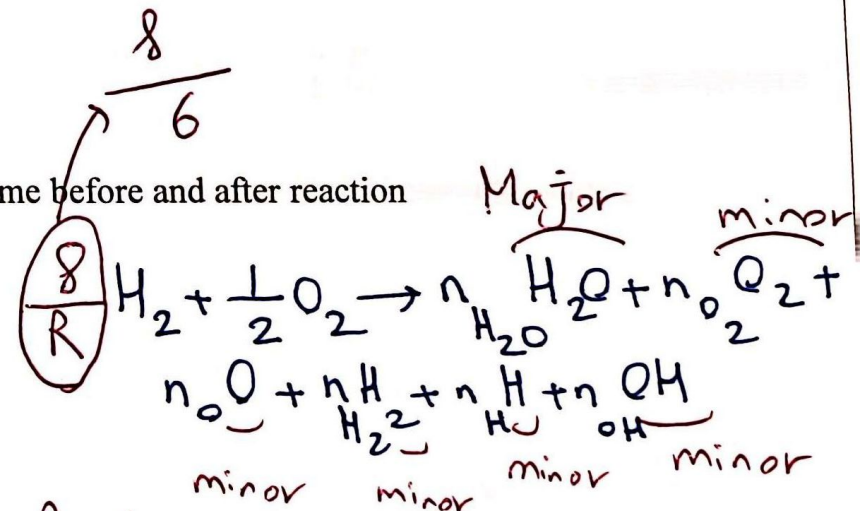
$$\text{if } R = R_{\text{sto}} \rightarrow \frac{8}{8} = 1$$

for our case $R = 6$

$$\frac{32 \times 1/2}{2N_{\text{H}_2}} = R$$

;

$$N_{\text{H}_2} = \frac{8}{R}$$



▪ The list of candidate reaction products is arrived at from our background knowledge of what species actually do occur.